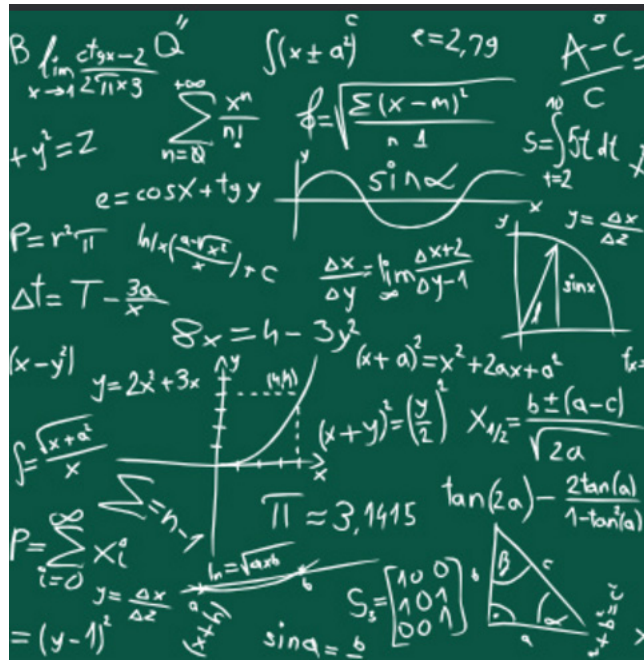


Numerical series



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Numerical series

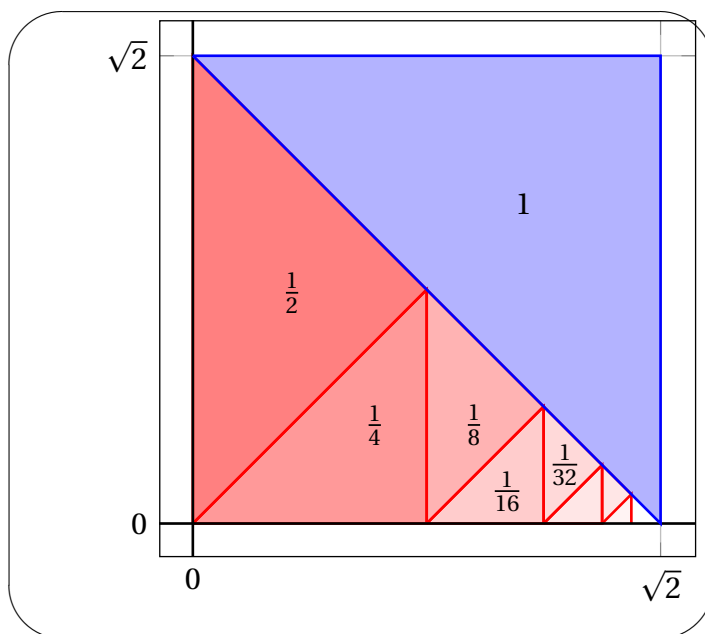
- We now introduce numerical series, which extend the operation of addition to an infinite number of terms.

The sum of infinitely many terms, even if they are all positive, can give a finite result.

- Imagine measuring a square of area 2 using the following procedure: we divide the square in half along the diagonal and measure the first right triangle: we obtain 1; then we divide the second right triangle in half and measure the first of the two resulting right triangles: we obtain $\frac{1}{2}$; the remaining right triangle is again divided in half... and so on indefinitely. We obtain the infinite sum:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots + \frac{1}{2^k} + \dots = \sum_{k=0}^{\infty} \frac{1}{2^k}$$

which, by the way it was constructed, must give 2 as its result.



Definition 1: of numerical series

Given a sequence $\{a_k\}_{k \in \mathbb{N}}$, we call **numerical series** of the terms a_k the expression:

$$\sum_{k=0}^{\infty} a_k$$

- It is read “series (or also sum) for k from 0 to $+\infty$ of a_k ”. The values a_k are called the **general terms of the series**.

Definition 2: of sequence of partial sums

The numbers

$$s_n = \sum_{k=0}^n a_k = a_0 + a_1 + \dots + a_n, \quad \forall n \in \mathbb{N}$$

are called the **n -th partial sums** of the series and they define **the sequence $\{s_n\}$ of partial sums**.

The **behavior (character) of the series** is determined by the limit of the sequence $\{s_n\}$ as n tends to infinity. We say that a series is **convergent, divergent, irregular**, if the **sequence $\{s_n\}$ of partial sums** is **convergent, divergent or irregular**, respectively.

Definition 3: of sum of the series

If the sequence $\{s_n\}$ of partial sums is **convergent**, that is, if

$$s_n \rightarrow s \in \mathbb{R} \quad \text{as } n \rightarrow +\infty$$

we say that s is **the sum of the series**, and we write $\sum_{k=0}^{\infty} a_k = s$.

Hence, if the sequence $\{s_n\}$ is **convergent**, we have:

$$\sum_{k=0}^{\infty} a_k = \lim_{n \rightarrow +\infty} \sum_{k=0}^n a_k = \lim_{n \rightarrow +\infty} s_n = s$$

The series makes precise the idea of a sum of infinitely many terms, that is, we compute the limit as $n \rightarrow +\infty$ of the finite sum of the first n terms.

- If, instead of summing starting from 0, we start from an index $n_0 > 0$, we write $\sum_{k=n_0}^{\infty} a_k$
- To denote a generic numerical series without specifying the starting index n_0 we will use the symbol $\sum a_k$

A generic numerical series $\sum a_k$ always involves **two different sequences**:

1. the sequence $\{a_k\}$ of the **terms of the series**
2. the sequence $\{s_n\}$ of **partial sums**

Example 1: behavior of a series

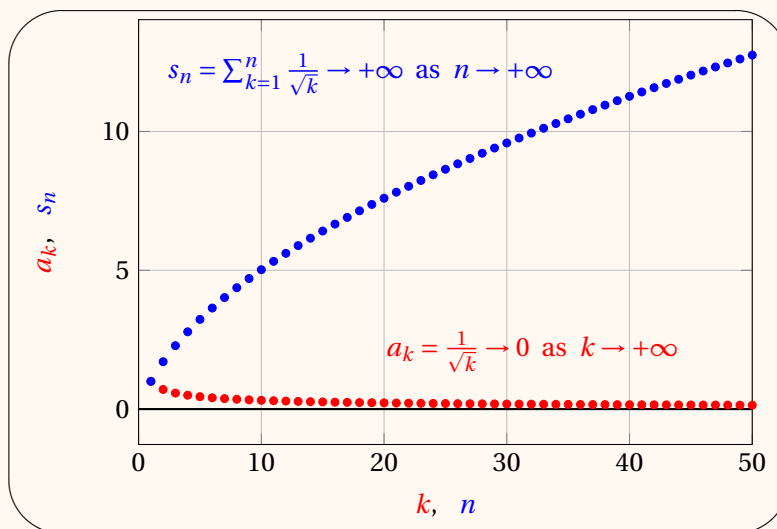
Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$$

We have:

$$s_n = \sum_{k=1}^n \frac{1}{\sqrt{k}} = 1 + \underbrace{\frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}}}_{n \text{ terms, each } \geq \frac{1}{\sqrt{n}}} \geq n \cdot \frac{1}{\sqrt{n}} = \sqrt{n} \rightarrow +\infty \text{ as } n \rightarrow +\infty$$

Hence the sequence of partial sums $\{s_n\}$ is divergent by the comparison theorem for sequences and, consequently, the series is divergent.



1.1 Main properties of numerical series

Remark 1

If a sequence $\{a_k\}$ has non-negative terms, that is, $a_k \geq 0, \forall k$, then the sequence of partial sums $\{s_n\}$ is increasing and regular.

Proof. Given a sequence $\{a_k\}$ with non-negative terms, the sequence of partial sums $\{s_n\}$ is increasing since:

$$s_{n+1} = s_n + \underbrace{a_{n+1}}_{\geq 0} \geq s_n, \quad \forall n, \quad \text{hence} \quad \lim_{n \rightarrow +\infty} s_n = \sup_{n \in \mathbb{N}} \{s_n\}$$

by the monotonicity theorem for sequences. Consequently the sequence $\{s_n\}$ cannot be irregular. It is therefore regular, that is, it either converges or diverges. $\square$

Given a sequence $\{a_k\}$ with non-negative terms:

- If $\{s_n\}$ is **increasing and bounded**, then it has a finite limit and $\sum a_k$ converges.
- If $\{s_n\}$ is **increasing and unbounded**, then $\sum a_k$ diverges to $+\infty$.

- This remark also holds for sequences with eventually non-negative terms, positive terms, or eventually positive terms.

Theorem 1

If a series $\sum a_k$ is convergent, then $\lim_{k \rightarrow +\infty} a_k = 0$

Proof. By definition of convergent series, the sequence of partial sums $\{s_n\}$ converges to a real number s , i.e., $s_n \rightarrow s \in \mathbb{R}$ as $n \rightarrow +\infty$. Without loss of generality we consider $n_0 = 0$.

We observe that the sequence $\{s_n\}$ can be defined recursively:

$$s_0 = a_0, \quad s_n = s_{n-1} + a_n, \quad \forall n \geq 1$$

Consequently we have:

$$a_n = s_n - s_{n-1} \quad \text{and hence} \quad \lim_{n \rightarrow +\infty} a_n = \lim_{n \rightarrow +\infty} (s_n - s_{n-1}) = s - s = 0$$

$\square$

The convergence of a series therefore implies that $a_k \rightarrow 0$ as $k \rightarrow +\infty$, that is:

$$\sum a_k \text{ convergent} \Rightarrow \lim_{k \rightarrow +\infty} a_k = 0 \quad (1)$$

but the converse is not true (Example 1 is a counterexample):

$$\lim_{k \rightarrow +\infty} a_k = 0 \not\Rightarrow \sum a_k \text{ convergent} \quad (2)$$

Hence the fact that $a_k \rightarrow 0$ as $k \rightarrow +\infty$ is a necessary but not sufficient condition for the convergence of a series.

- From the contrapositive of (1) we have:

$$\lim_{k \rightarrow +\infty} a_k \neq 0 \Rightarrow \sum a_k \text{ is divergent or irregular}$$

that is, if the limit is not equal to zero, then the series is not convergent.

1.2 Tails of series

If we modify a finite number of terms of a series, the value of the sum may change, but the behavior of the series remains unchanged.

- If we modify the value of one term, for example a_{n_0} , for $n \geq n_0$ the sequences of partial sums, the original and the modified one, differ only by that term: hence both converge, both diverge, or both are irregular. The same holds if we modify a finite number of terms.

The results on the behavior of series therefore also hold if the hypotheses are satisfied “eventually”, i.e., from a certain index n_0 onward.

Definition 4: of tail of a series

Given a series $\sum_{k=0}^{\infty} a_k$ and a value $m \in \mathbb{N}$, the series $\sum_{k=m+1}^{\infty} a_k$ is called a **tail of the series**.

- Given a series $\sum_{k=0}^{\infty} a_k$ and $m \in \mathbb{N}$ we have:

$$\sum_{k=m+1}^n a_k = \left(\sum_{k=0}^n a_k \right) - \left(\sum_{k=0}^m a_k \right) = s_n - s_m, \quad \forall n > m \quad (3)$$

hence for every tail (that is, for every m) the associated sequence of partial sums is equal to that of the original series minus a constant. Consequently we have the following remark:

Every tail of a series has the same behavior as the original series.

- Given a convergent series, taking the limit as $n \rightarrow +\infty$ in (3) we obtain the relation:

$$\sum_{k=m+1}^{\infty} a_k = \left(\sum_{k=0}^{\infty} a_k \right) - \left(\sum_{k=0}^m a_k \right) = s - s_m \quad \text{as } n \rightarrow +\infty \quad (4)$$

Consequently we have the following remark:

Given $m \in \mathbb{N}$, every tail $\sum_{k=m+1}^{\infty} a_k$ of a convergent series can be interpreted as the **error** that we make when approximating the sum s with the partial sum s_m .

Theorem 2

If a series $\sum a_k$ is convergent, then

$$\underbrace{s - s_m}_{= \sum_{k=m+1}^{\infty} a_k} \rightarrow 0 \quad \text{as } m \rightarrow +\infty$$

Proof. For every $n > m$, the n -th partial sum of the series is:

$$s_n = s_m + \sum_{k=m+1}^n a_k$$

Taking the limit as $n \rightarrow +\infty$, we have:

$$\begin{aligned} s &= s_m + \lim_{n \rightarrow +\infty} \sum_{k=m+1}^n a_k \\ &= \sum_{k=m+1}^{\infty} a_k \end{aligned}$$

Now taking the limit as $m \rightarrow +\infty$, we have:

$$s = s + \lim_{m \rightarrow +\infty} \sum_{k=m+1}^{\infty} a_k \Rightarrow \lim_{m \rightarrow +\infty} \sum_{k=m+1}^{\infty} a_k = 0$$

that is, the claim of the theorem. □

This theorem tells us that, if a series is convergent, then the value $s - s_m$, that is, the error that we make when approximating the sum s with the partial sum s_m , tends to zero as $m \rightarrow +\infty$. In other words, the tail of a convergent series tends to zero as m tends to infinity.

1.3 Harmonic series

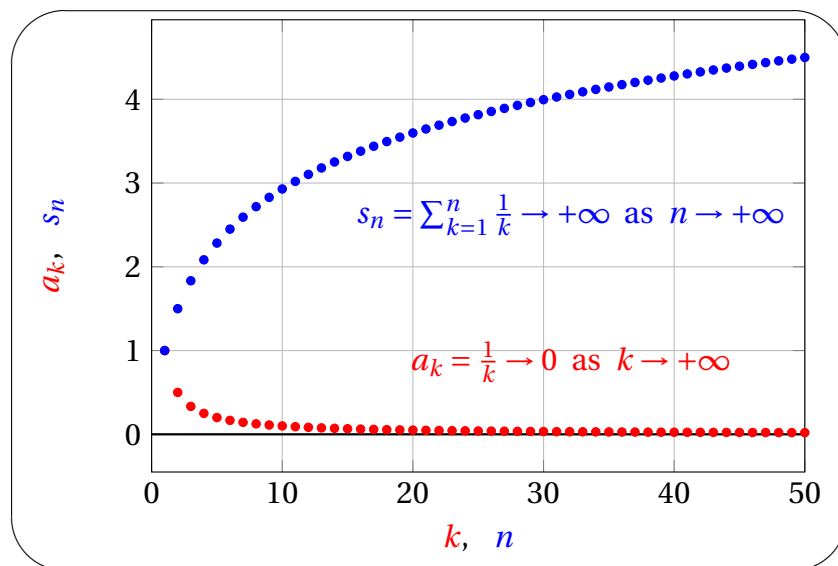
Definition 5: of harmonic series

The **harmonic series** is the series $\sum_{k=1}^{\infty} \frac{1}{k}$

Theorem 3: behavior of the harmonic series

The harmonic series diverges to $+\infty$

- Graphically we have:



The harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ is a counterexample that proves (2), since:

$$\lim_{k \rightarrow +\infty} \frac{1}{k} = 0 \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges to } +\infty$$

- Before proving the theorem, we observe that for certain values of n the partial sums are:

$$\begin{aligned}
\underbrace{s_1}_{=s_{2^0}} &= 1 + \frac{0}{2}, & \underbrace{s_2}_{=s_{2^1}} &= s_1 + \frac{1}{2} = 1 + \frac{0}{2} + \frac{1}{2} = 1 + \frac{1}{2}, & \underbrace{s_4}_{=s_{2^2}} &= s_2 + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{\geq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}} \geq 1 + \frac{1}{2} + \frac{1}{2} = 1 + \frac{2}{2} \\
\underbrace{s_8}_{=s_{2^3}} &= s_4 + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{\geq \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}} \geq 1 + \frac{2}{2} + \frac{1}{2} = 1 + \frac{3}{2} \\
\underbrace{s_{16}}_{=s_{2^4}} &= s_8 + \underbrace{\left(\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16}\right)}_{\geq \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{1}{2}} \geq 1 + \frac{3}{2} + \frac{1}{2} = 1 + \frac{4}{2}
\end{aligned}$$

Proof. We prove by induction on 2^n that

$$s_{2^n} \geq 1 + \frac{n}{2}, \quad \forall n \in \mathbb{N}$$

Base case. Let $n = 0$. Then the statement becomes $s_{2^0} \geq 1 + \frac{0}{2}$, i.e., $1 \geq 1$, which is clearly true.

Inductive step. Assume that it is true for 2^{n-1} and let us prove it for 2^n . By the inductive hypothesis, we have:

$$s_{2^{n-1}} \geq 1 + \frac{n-1}{2}$$

Moreover, we have:

$$\begin{aligned}
s_{2^n} &= s_{2^{n-1}} + \underbrace{\sum_{k=2^{n-1}+1}^{2^n} \left(\frac{1}{k}\right)}_{\geq 2^{n-1} \cdot \frac{1}{2^n} = \frac{1}{2}} \quad \text{hence} \quad s_{2^n} \geq 1 + \frac{n-1}{2} + \frac{1}{2} = 1 + \frac{n}{2}
\end{aligned}$$

which is exactly the desired statement for 2^n .

We therefore have:

$$s_{2^n} \geq 1 + \frac{n}{2} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty$$

and $\{s_n\}$ is unbounded above and also monotonically increasing, since $a_k = 1/k > 0, \forall k > 1$. Consequently $\{s_n\}$ diverges to $+\infty$ by the monotonicity theorem for sequences, and the harmonic series diverges to $+\infty$.

□

The sequence $a_k = 1/k, \forall k \geq 1$, of the harmonic series has positive terms, hence the sequence of partial sums $\{s_n\}$ is increasing, and it is divergent since it is unbounded.

1.4 Geometric series

Definition 6: of geometric series

Given $q \in \mathbb{R}$, the **geometric series** with **ratio** q is the series $\sum_{k=0}^{\infty} q^k$

Theorem 4: behavior and sum of the geometric series

Given the ratio $q \in \mathbb{R}$,

$$\text{the geometric series } \sum_{k=0}^{\infty} q^k \text{ is } \begin{cases} \text{convergent} & \text{if } |q| < 1 \\ \text{divergent to } +\infty & \text{if } q \geq 1 \\ \text{irregular} & \text{if } q \leq -1 \end{cases}$$

If the geometric series is convergent, its sum s is $\frac{1}{1-q}$

Proof. Given $q \in \mathbb{R}$ and $n \in \mathbb{N}$, the n -th partial sum of the geometric series is:

$$\sum_{k=0}^n q^k = \begin{cases} \frac{1 - q^{n+1}}{1 - q} & \text{if } q \neq 1 \\ n + 1 & \text{if } q = 1 \end{cases}$$

since it equals the sum of the first $n + 1$ terms of the geometric progression. Moreover, we have:

$$\lim_{n \rightarrow +\infty} q^n = \begin{cases} +\infty & \text{if } q > 1 \\ 1 & \text{if } q = 1 \\ 0 & \text{if } |q| < 1 \\ \text{does not exist} & \text{if } q \leq -1 \end{cases}$$

Hence, if $q \neq 1$:

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \frac{1 - q^{n+1}}{1 - q} = \frac{1}{1 - q} \lim_{n \rightarrow +\infty} (1 - q^{n+1}) = \begin{cases} \frac{1}{1 - q} & \text{if } |q| < 1 \\ +\infty & \text{if } q > 1 \\ \text{does not exist} & \text{if } q \leq -1 \end{cases}$$

and if $q = 1$ we have:

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} (n + 1) = +\infty$$

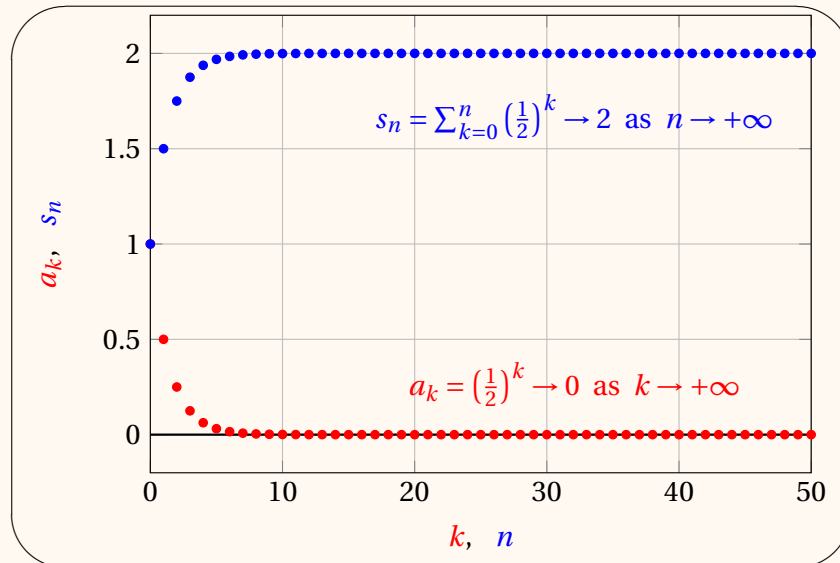
□

Example 2: geometric series

Let us determine the behavior of the series:

$$\sum_{k=0}^{\infty} \frac{1}{2^k} = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k$$

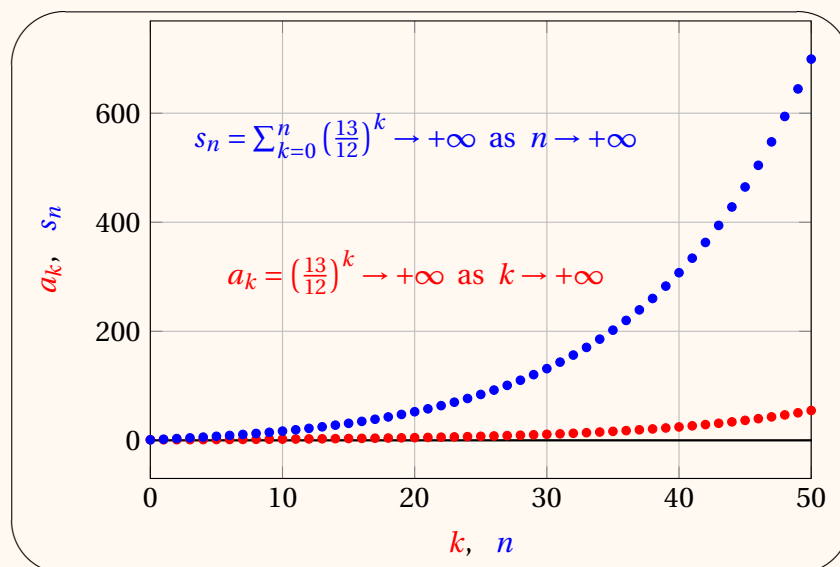
It is a geometric series with ratio $q = 1/2$, hence it is convergent and its sum s is $1/(1 - 1/2) = 2$.



Let us determine the behavior of the series:

$$\sum_{k=0}^{\infty} \left(\frac{13}{12}\right)^k$$

It is a geometric series with ratio $q = 13/12$, hence it is divergent.



1.5 Telescoping series

Definition 7: of telescoping series

A **telescoping series** is a series of the form:

$$\sum_{k=n_0}^{\infty} (b_k - b_{k+1}) \quad \text{where } \{b_k\} \text{ is a sequence}$$

Theorem 5: behavior and sum of the telescoping series

A telescoping series converges, diverges or is irregular according to whether the sequence $\{b_k\}$ converges, diverges or is irregular, respectively.

$$\text{If } \lim_{k \rightarrow +\infty} b_k = \ell \in \mathbb{R}^* \quad \text{then} \quad \sum_{k=n_0}^{\infty} (b_k - b_{k+1}) = b_{n_0} - \underbrace{\lim_{k \rightarrow +\infty} b_k}_{=\ell \in \mathbb{R}^*}$$

Moreover, if $\ell \in \mathbb{R}$, then the telescoping series is convergent and its sum s is $b_{n_0} - \ell$

Proof. We have

$$\begin{aligned} s_n &= \sum_{k=n_0}^{n_0+n} (b_k - b_{k+1}) \\ &= (b_{n_0} - b_{n_0+1}) + (b_{n_0+1} - b_{n_0+2}) + \dots + (b_{n_0+n} - b_{n_0+n+1}) \\ &= b_{n_0} - b_{n_0+n+1} \end{aligned}$$

hence:

$$\sum_{k=n_0}^{\infty} (b_k - b_{k+1}) = \lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} (b_{n_0} - b_{n_0+n+1}) = b_{n_0} - \lim_{n \rightarrow +\infty} b_n$$

□

Remark 2

The series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$, called **Mengoli's series**, is convergent and its sum s is equal to 1.

Proof. We have:

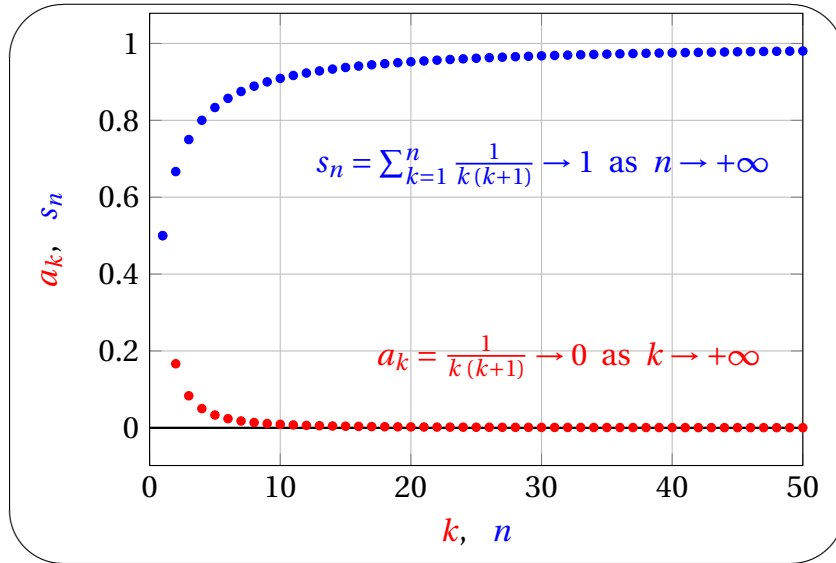
$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \frac{(k+1) - k}{k(k+1)} = \sum_{k=1}^{\infty} \underbrace{\frac{1}{k}}_{=b_k} - \underbrace{\frac{1}{k+1}}_{=b_{k+1}}$$

Mengoli's series therefore has the form:

$$b_k - b_{k+1} \quad \text{with} \quad b_k = \frac{1}{k} \quad \text{and moreover} \quad \lim_{k \rightarrow +\infty} b_k = \lim_{k \rightarrow +\infty} \frac{1}{k} = 0$$

hence it is a convergent telescoping series ($n_0 = 1$) and moreover $s = b_1 = 1$. $\square$

- Graphically we have:



Proof. Alternative proof

For every $n \geq 1$, the value of the partial sum of Mengoli's series is:

$$s_n = \sum_{k=1}^n \frac{1}{k(k+1)} = \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{n+1}$$

Hence we have

$$\sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} 1 - \frac{1}{n+1} = 1$$

Consequently Mengoli's series is convergent and its sum s is 1. $\square$

Example 3: telescoping series

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{1}{(3k+2)(3k+5)}$$

We have:

$$\begin{aligned} \frac{1}{(3k+2)(3k+5)} &= \frac{\alpha}{3k+2} - \frac{\beta}{3k+5} = \frac{\alpha(3k+5) - \beta(3k+2)}{(3k+2)(3k+5)} = \frac{3(\alpha - \beta)k + 5\alpha - 2\beta}{(3k+2)(3k+5)} \\ &\Rightarrow \begin{cases} 3(\alpha - \beta) = 0 \\ 5\alpha - 2\beta = 1 \end{cases} \Rightarrow \alpha = \beta, \quad 3\alpha = 1 \text{ hence } \alpha = \beta = \frac{1}{3} \end{aligned}$$

consequently:

$$\frac{1}{(3k+2)(3k+5)} = \frac{1}{3} \left(\frac{1}{3k+2} - \frac{1}{3k+5} \right) = \frac{1}{3} \left(\underbrace{\frac{1}{3k+2}}_{=b_k} - \underbrace{\frac{1}{3(k+1)+2}}_{=b_{k+1}} \right)$$

The series therefore has the form:

$$b_k - b_{k+1} \quad \text{with} \quad b_k = \frac{1}{3k+2} \quad \text{and moreover} \quad \lim_{k \rightarrow +\infty} b_k = \lim_{k \rightarrow +\infty} \frac{1}{3k+2} = 0$$

that is, a telescoping series ($n_0 = 1$), and we have:

$$\sum_{k=1}^{\infty} \frac{1}{(3k+2)(3k+5)} = \frac{1}{3} \left(b_1 - \underbrace{\lim_{k \rightarrow +\infty} b_k}_{=0} \right) = \frac{1}{3} b_1 = \frac{1}{15}$$

Hence it is a convergent telescoping series and its sum s is $\frac{1}{15}$.